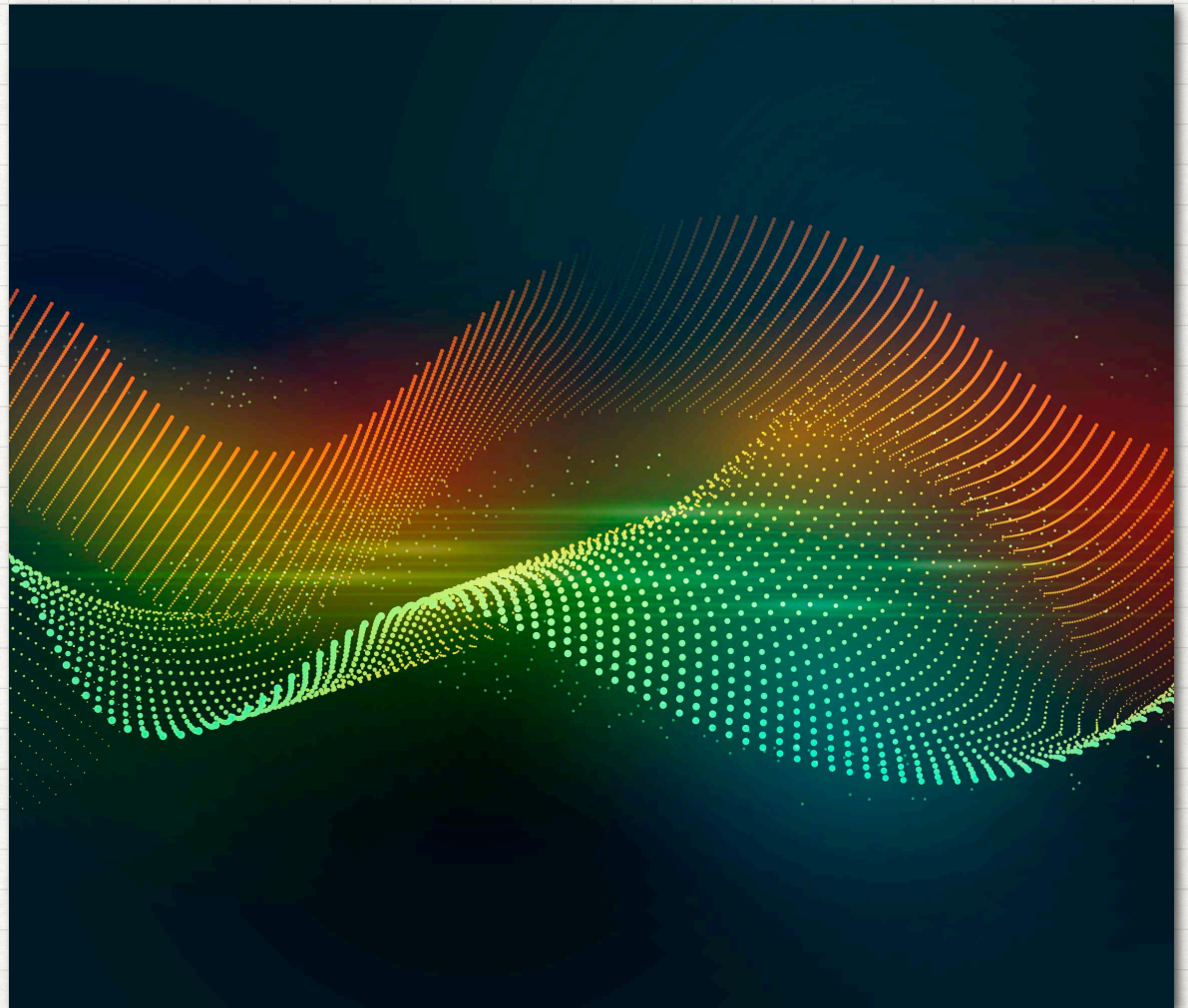


LECTURE 04
FOURIER
ANALYSIS OF
DISCRETE-TIME
SIGNALS



INTRODUCTION



Time



Frequency

- **Continuous-time Signals (t)**

- Periodic signals - $x(t)$
- Aperiodic signals - $x(t)$

$x(t)$

a_n

- **Discrete-time Signals [n]**

- Periodic signals - $x[n]$
- Aperiodic signals - $x[n]$
- DFT - $x[n]$

$x[n]$

a_n

$x[n]$

$X(e^{j\omega})$

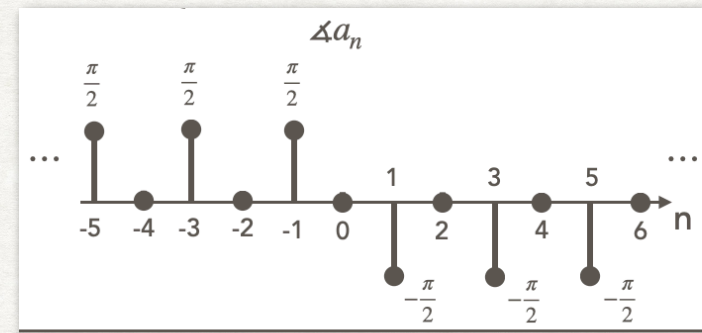
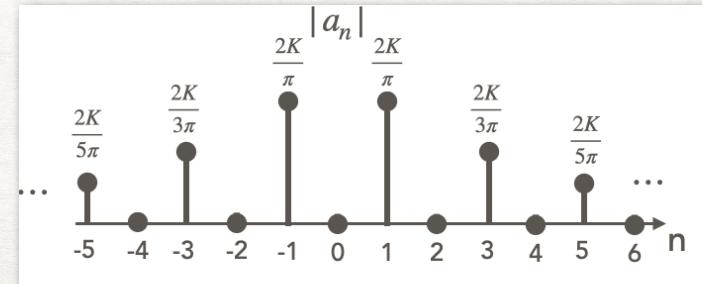
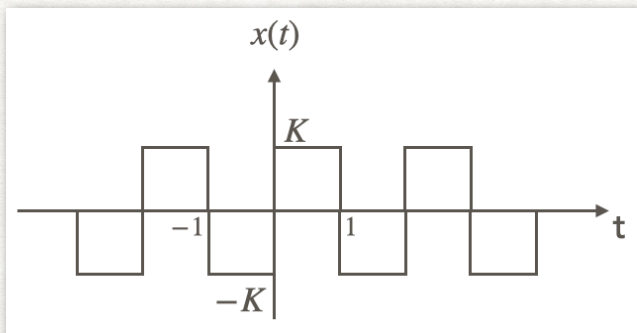
$x[n]$

$X[k]$

CT FOURIER SERIES

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$$

$$a_k = \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_0^T x(t) e^{-jk(2\pi/T)t} dt$$



OUTLINE

- Fourier series representation of discrete-time periodic signals
- Discrete-time Fourier transform
- Properties of Discrete-time Fourier transform
- DFT and its properties

FOURIER ANALYSIS OF DISCRETE-TIME SYSTEMS

A discrete-time signal $x[n]$ is periodic with period N if $x[n] = x[n + N]$

The complex exponential, $e^{j(2\pi/N)n}$ is periodic with period N , thus the set of all discrete-time complex expo signals with period N is given by

$$\phi_k[n] = e^{j(k\omega_0)n} = e^{jk(2\pi/N)n}, k = 0, \pm 1, \pm 2, \dots \quad (4.1)$$

All of these signals have fundamental frequencies that are multiples of $2\pi/N$ and thus are harmonically related.

Discrete-time complex exponentials which differ in frequency by a multiple of 2π are identical. Thus, $\phi_0[n] = \phi_N[n]$, $\phi_1[n] = \phi_{N+1}[n]$, in general

$$\phi_k[n] = \phi_{k+rN}[n]$$

DIFFER FROM THE
CONTINUOUS-TIME SYSTEM!

(4.2)

FOURIER ANALYSIS OF DISCRETE-TIME SYSTEMS

The representation of more periodic sequences in terms of linear combinations of the sequences $\phi_k[n]$ has a form

$$x[n] = \sum_k a_k \phi_k[n] = \sum_k a_k e^{jk(\omega_0)n} = \sum_k a_k e^{jk(2\pi/N)n} \quad (4.3)$$

Handwritten notes: A blue squiggle is above the first sum. A blue arrow points from the first sum to the second. A blue arrow points from the second sum to the third. A blue circle is around the exponential term in the second sum. A blue circle is around the exponential term in the third sum.

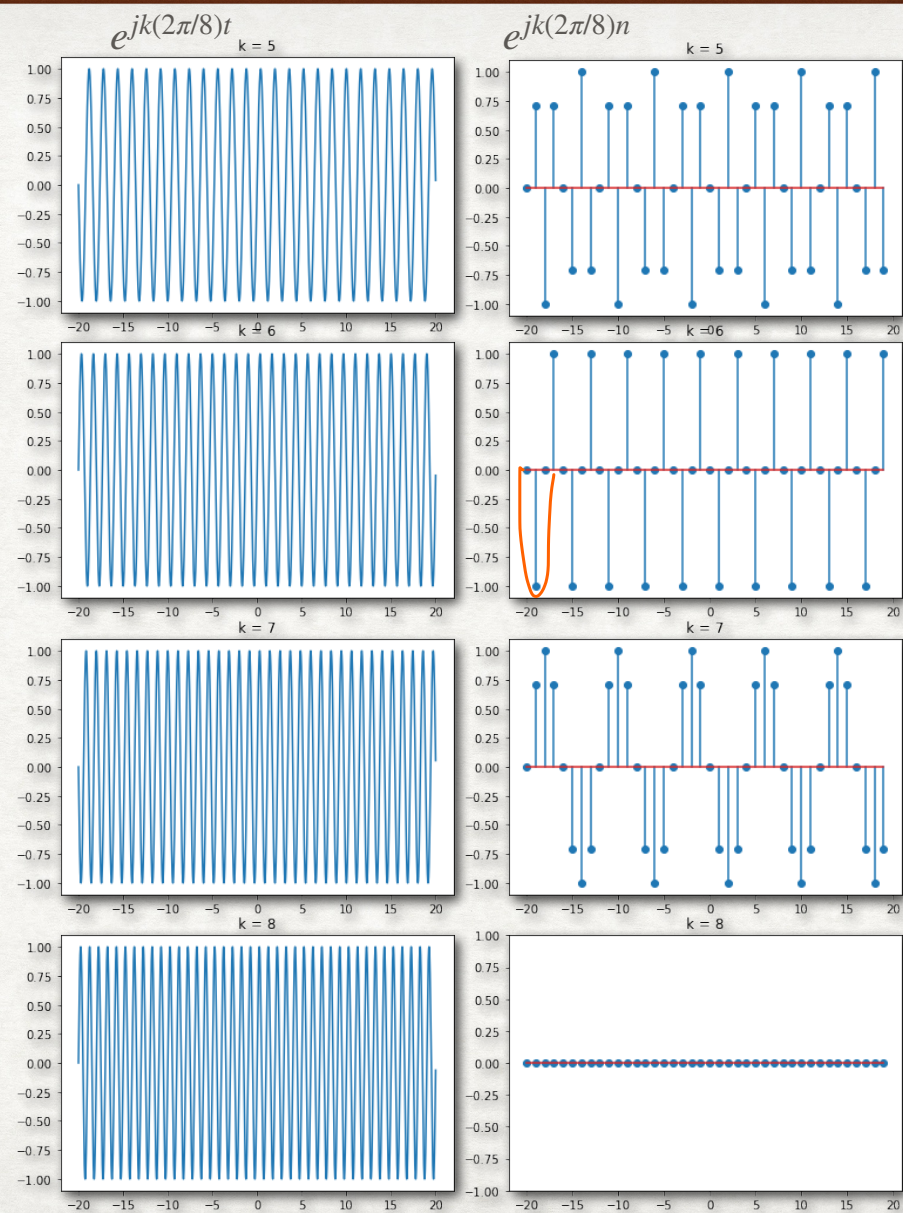
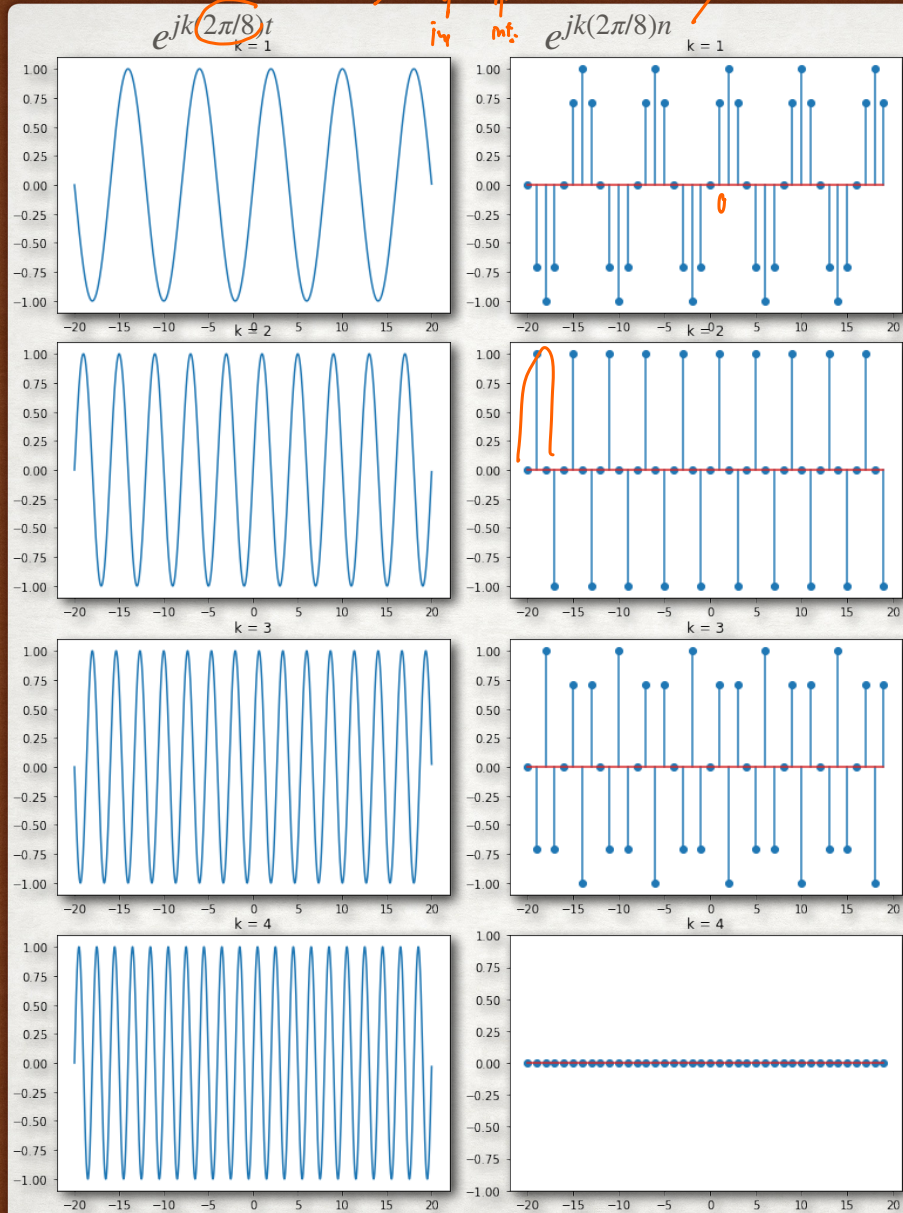
The sequence $\phi_k[n]$ are distinct only over a range of N successive values of k , the summation in (4.3) need only include terms over this range. We indicate this by expressing the limits of the summations as $k = \langle N \rangle$

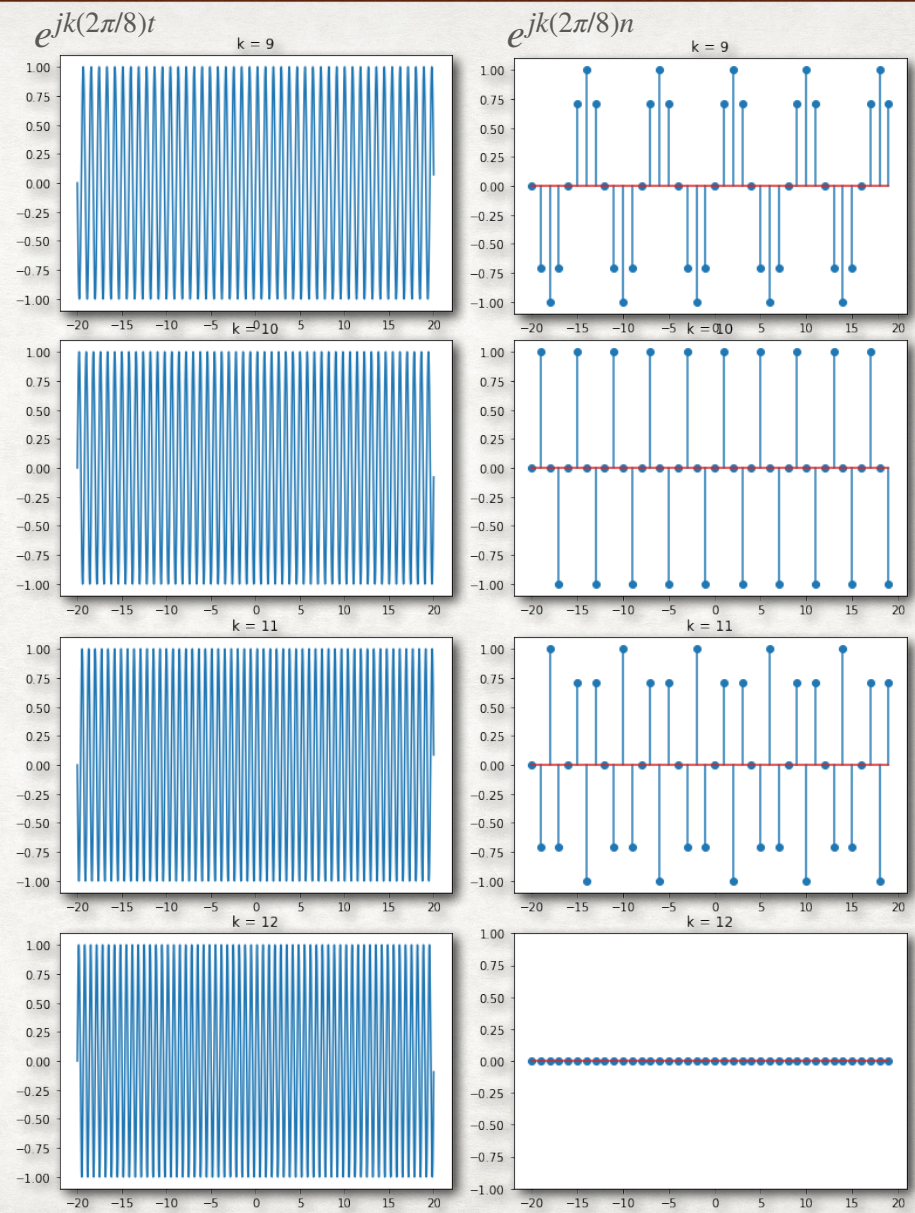
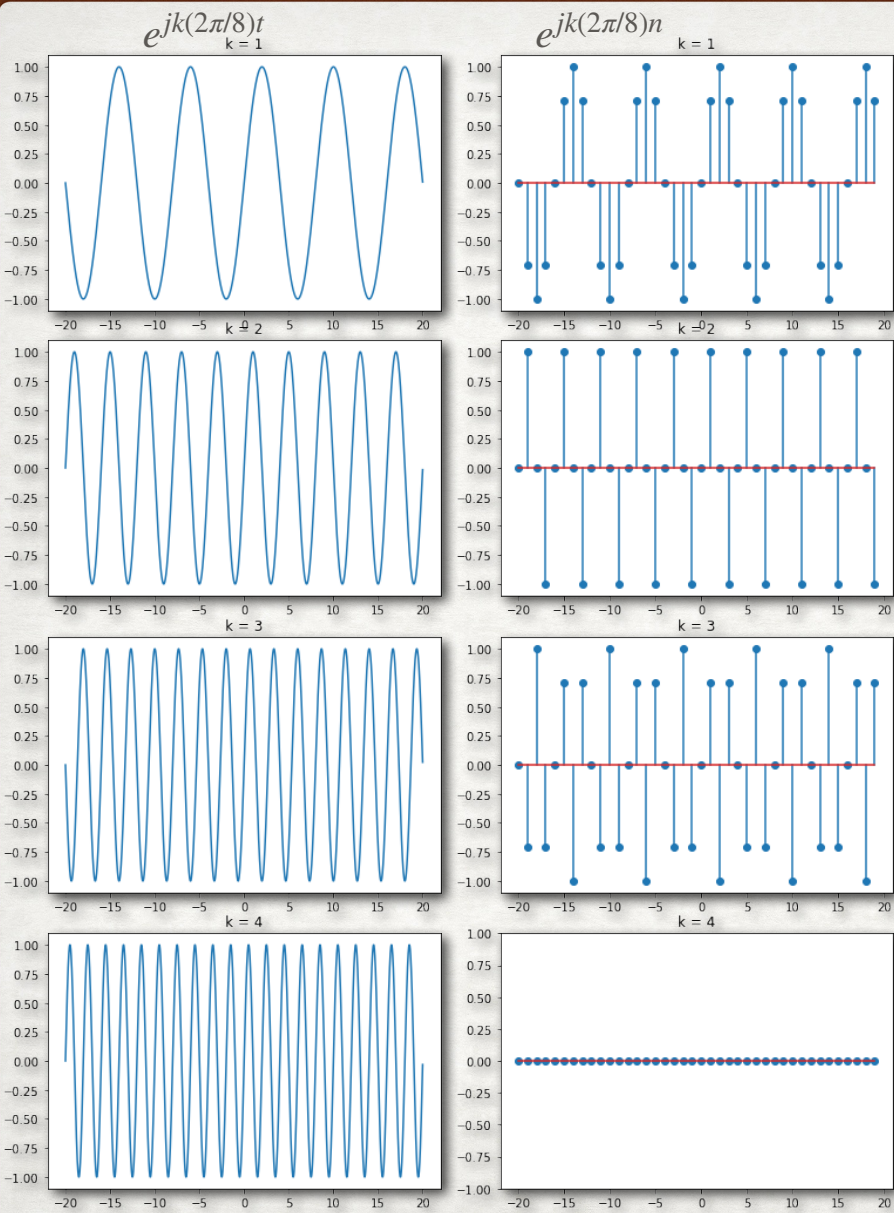
$$x[n] = \sum_{\langle N \rangle} a_k \phi_k[n] = \sum_{\langle N \rangle} a_k e^{jk(\omega_0)n} = \sum_{\langle N \rangle} a_k e^{jk(2\pi/N)n} \quad (4.4)$$

Handwritten notes: An orange arrow points to the first sum. An orange circle is around the first sum. An orange circle is around the exponential term in the second sum. An orange circle is around the exponential term in the third sum.

- You can take on the values $k = 0, 1, \dots, N-1$ or $k = 3, 4, \dots, N+2$
- (4.4) is referred as the **discrete-time Fourier series** and the coefficients a_k as the **Fourier series coefficients**

$\omega_0 = \frac{2\pi}{8}$
 $\cos(2\pi f_0 n)$
 $\frac{1}{8} \cdot N = m$
 $f_0 N = m$
 $m = 1$ ($N=8$)
 $\cos(k \frac{2\pi}{8} n) + j \sin(k \frac{2\pi}{8} n)$

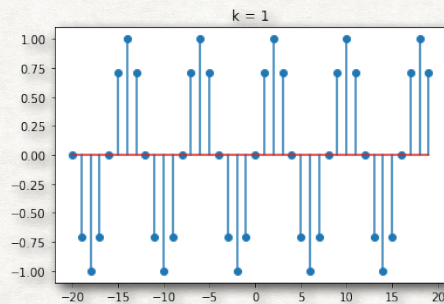




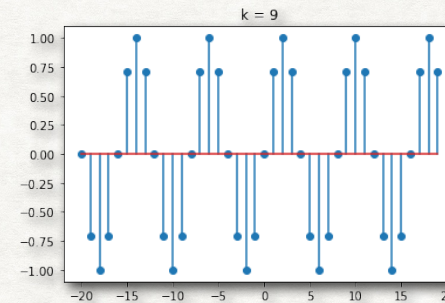
$k=1$ 17, 25

$k + \textcircled{N}$
 $k+8$ $k=9$

$$e^{j\frac{2\pi}{8}n} = \cos\left(\frac{2\pi}{8}n\right) + j \sin\left(\frac{2\pi}{8}n\right)$$

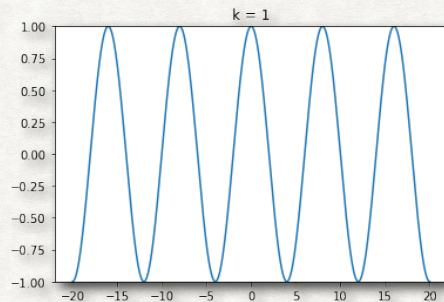


$$e^{j9 \cdot \frac{2\pi}{8}n} = \cos\left(9 \cdot \frac{2\pi}{8}n\right) + j \sin\left(9 \cdot \frac{2\pi}{8}n\right)$$



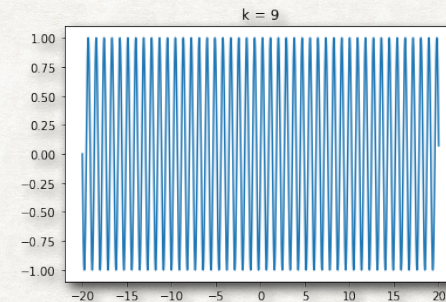
Highest frequency is at π

$$\cos\left(\frac{2\pi}{8}n\right) + j \sin\left(\frac{2\pi}{8}n\right) = \cos\left(9 \cdot \frac{2\pi}{8}n\right) + j \sin\left(9 \cdot \frac{2\pi}{8}n\right)$$



$e^{j9 \cdot \frac{2\pi}{8}t}$ has higher frequency than $e^{j\frac{2\pi}{8}t}$ for 9 times

$$e^{j\frac{2\pi}{8}t} = \cos\left(\frac{2\pi}{8}t\right) + j \sin\left(\frac{2\pi}{8}t\right)$$

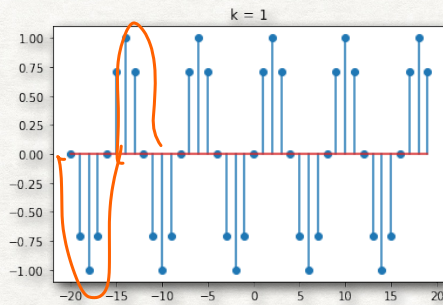


Highest frequency can reach ∞

$$e^{j9 \cdot \frac{2\pi}{8}t} = \cos\left(9 \cdot \frac{2\pi}{8}t\right) + j \sin\left(9 \cdot \frac{2\pi}{8}t\right)$$

$k=1$

$$e^{j\frac{2\pi}{8}n} = \cos\left(\frac{2\pi}{8}n\right) + j \sin\left(\frac{2\pi}{8}n\right)$$

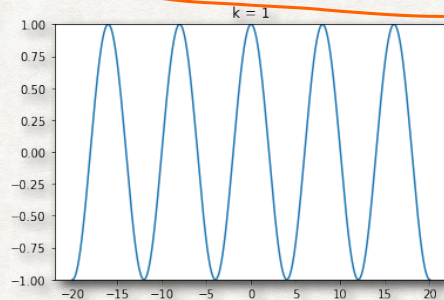


$$\begin{aligned} \cos\left(7 \cdot \frac{2\pi}{8}n\right) &= \cos\left(\overset{N-1}{(8-1)} \frac{2\pi}{8}n\right) \\ &= \cos\left(-\frac{2\pi}{8}n\right) = \cos\left(\frac{2\pi}{8}n\right) \end{aligned}$$

$$\sin\left(7 \cdot \frac{2\pi}{8}n\right) = \sin\left(-\frac{2\pi}{8}n\right) = -\sin\left(\frac{2\pi}{8}n\right)$$

$$e^{j7 \cdot \frac{2\pi}{8}n} = \cos\left(7 \cdot \frac{2\pi}{8}n\right) + j \sin\left(7 \cdot \frac{2\pi}{8}n\right) = \cos\left(\frac{2\pi}{8}n\right) - j \sin\left(\frac{2\pi}{8}n\right)$$

Same magnitude, opposite phase

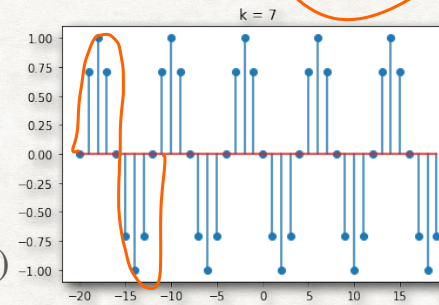


$e^{j7 \cdot \frac{2\pi}{8}t}$ has higher frequency than $e^{j\frac{2\pi}{8}t}$ for 7 times

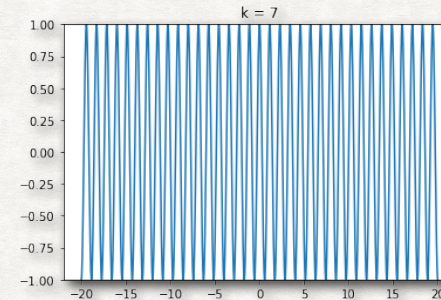
$$e^{j\frac{2\pi}{8}t} = \cos\left(\frac{2\pi}{8}t\right) + j \sin\left(\frac{2\pi}{8}t\right)$$

$k=7$

$$e^{j7 \cdot \frac{2\pi}{8}n} = \cos\left(7 \cdot \frac{2\pi}{8}n\right) + j \sin\left(7 \cdot \frac{2\pi}{8}n\right)$$



Highest frequency is at π



Highest frequency can reach ∞

$$e^{j7 \cdot \frac{2\pi}{8}t} = \cos\left(7 \cdot \frac{2\pi}{8}t\right) + j \sin\left(7 \cdot \frac{2\pi}{8}t\right)$$

$$x[n] = \sum_{k=-\infty}^{\infty} a_k e^{jk \frac{2\pi}{N} \cdot n}$$

FOURIER ANALYSIS OF DISCRETE-TIME SYSTEMS

From (4.4), multiplying both sides by $e^{-jr(2\pi/N)n}$ and summing over N terms, we obtain

$$\sum_{n=-\infty}^{\infty} x[n] e^{-jr(2\pi/N)n} = \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/N)n} e^{-jr(2\pi/N)n} \right] \quad (4.5)$$

• Interchanging order of the summation on the right-hand side, we have

$$\sum_{n=-\infty}^{\infty} x[n] e^{-jr(2\pi/N)n} = \sum_{k=-\infty}^{\infty} a_k \sum_{n=-\infty}^{\infty} e^{jk(2\pi/N)n} e^{-jr(2\pi/N)n} \quad (4.6)$$

$k \neq r \rightarrow 0$
 $k = r \rightarrow N$

The term $\sum_{n=-\infty}^{\infty} e^{jk(2\pi/N)n} e^{-jr(2\pi/N)n} = \sum_{n=-\infty}^{\infty} e^{j(k-r)(2\pi/N)n}$ using geometric series, we obtain

$$\sum_{n=-\infty}^{\infty} e^{j(k-r)(2\pi/N)n} = \begin{cases} N, & (k-r) = 0, \pm N, \pm 2N, \dots \\ 0, & \text{otherwise} \end{cases} \quad (4.7)$$

$a_r N$

Geometric Series

$$a + ar + ar^2 + \dots + ar^N = \sum_{k=0}^N ar^k = a \left(\frac{1-r^{N+1}}{1-r} \right)$$

$$a + ar + ar^2 + \dots = \sum_{k=0}^{\infty} ar^k = a \left(\frac{1}{1-r} \right), \text{ for } |r| < 1$$

$$a_r = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jr(2\pi/N)n}$$

$$e^{j(0) \cdot 2\pi} = 1$$

$$\sum_{n=0}^{N-1} \left(e^{j(k-r) \frac{2\pi}{N} n} \right) = ?$$

$$\sum_{k=0}^N ar^k = a \left(\frac{1-r^{N+1}}{1-r} \right)$$

$$r \rightarrow e^{j(k-r) \frac{2\pi}{N}}$$

$$a = 1$$

$$k=r \quad \sum_{n=0}^{N-1} 1 = N$$

$$\sum_{n=0}^{N-1} 1$$

$$= \frac{1 - e^{j(k-r) \frac{2\pi}{N} N}}{1 - e^{j(k-r) \frac{2\pi}{N}}}$$

$$k \neq r = \frac{1 - e^{j(k-r) 2\pi}}{1 - e^{j(k-r) \frac{2\pi}{N}}}$$

$$= 0$$

FOURIER ANALYSIS OF DISCRETE-TIME SYSTEMS



For $k = r$, left-term in (4.7) equals N , the (4.6) reduces to

$$a_r = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jr(2\pi/N)n} \quad (4.8)$$

In summary, discrete-time Fourier series pair:

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(\omega_0)n} = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n} \quad (4.9)$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n} \quad (4.10)$$

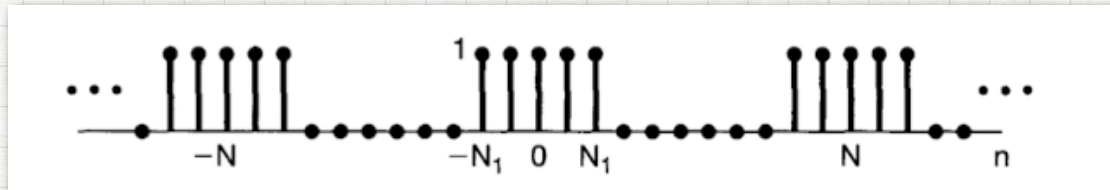
DISCRETE-TIME FOURIER SERIES

EXERCISE #1



Considering a periodic square wave, find Fourier series of the signal.

$$x[n] = 1 \text{ for } -N_1 \leq n \leq N_1$$



Plot Fourier coefficients for $2N_1 + 1 = 5$
and $N = 10, 20, 40$

Let $m = n + N_1$

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{m=0}^{2N_1} e^{-jk(2\pi/N)(m-N_1)} \\ &= \frac{1}{N} e^{jk(2\pi/N)N_1} \sum_{m=0}^{2N_1} e^{-jk(2\pi/N)m} \end{aligned}$$

$$\begin{aligned} a_k &= \frac{1}{N} e^{jk(2\pi/N)N_1} \left(\frac{1 - e^{-jk2\pi(2N_1+1)/N}}{1 - e^{-jk(2\pi/N)}} \right) \\ &= \frac{1}{N} \frac{e^{-jk(2\pi/2N)} [e^{jk2\pi(N_1+1/2)/N} - e^{-jk2\pi(N_1+1/2)/N}]}{e^{-jk(2\pi/2N)} [e^{jk(2\pi/2N)} - e^{-jk(2\pi/2N)}]} \\ &= \frac{1}{N} \frac{\sin[2\pi k(N_1 + 1/2)/N]}{\sin(\pi k/N)}, \quad k \neq 0, \pm N, \pm 2N, \dots \end{aligned}$$

and
$$a_k = \frac{2N_1 + 1}{N}, \quad k = 0, \pm N, \pm 2N, \dots$$

DISCRETE-TIME FOURIER SERIES

EXERCISE #1



Considering a periodic square wave, find Fourier series of the signal.

$$x[n] = 1 \text{ for } -N_1 \leq n \leq N_1$$

Plot Fourier coefficients for $2N_1 + 1 = 5$
and $N = 10, 20, 40$

$$\begin{aligned} a_k &= \frac{1}{N} e^{jk(2\pi/N)N_1} \left(\frac{1 - e^{-jk2\pi(2N_1+1)/N}}{1 - e^{-jk(2\pi/N)}} \right) \\ &= \frac{1}{N} \frac{e^{-jk(2\pi/2N)} [e^{jk2\pi(N_1+1/2)/N} - e^{-jk2\pi(N_1+1/2)/N}]}{e^{-jk(2\pi/2N)} [e^{jk(2\pi/2N)} - e^{-jk(2\pi/2N)}]} \\ &= \frac{1}{N} \frac{\sin[2\pi k(N_1 + 1/2)/N]}{\sin(\pi k/N)}, \quad k \neq 0, \pm N, \pm 2N, \dots \end{aligned}$$

$$a_k = \frac{2N_1 + 1}{N}, \quad k = 0, \pm N, \pm 2N, \dots$$

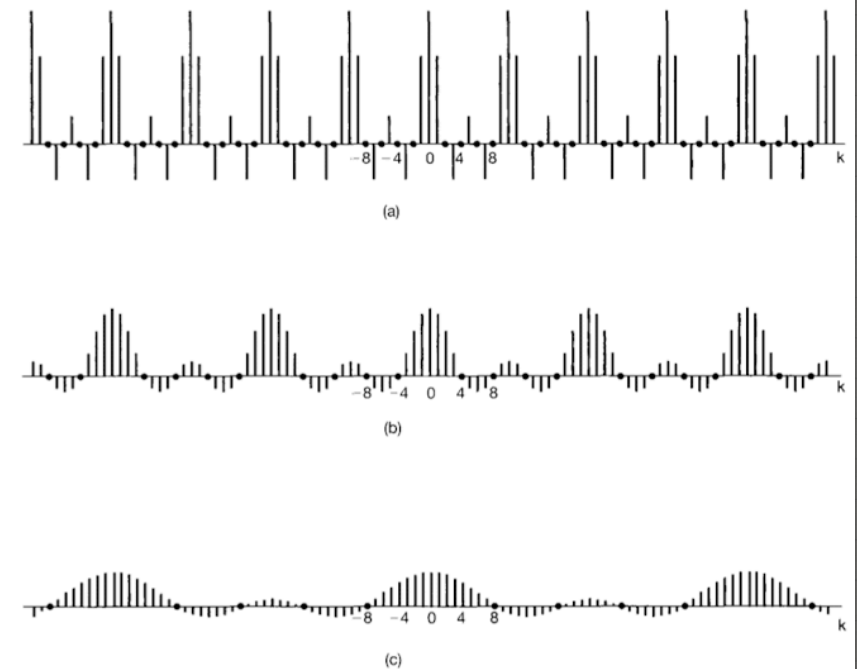
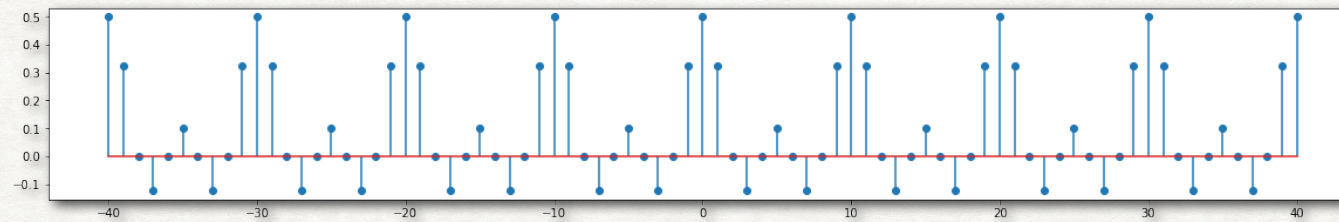


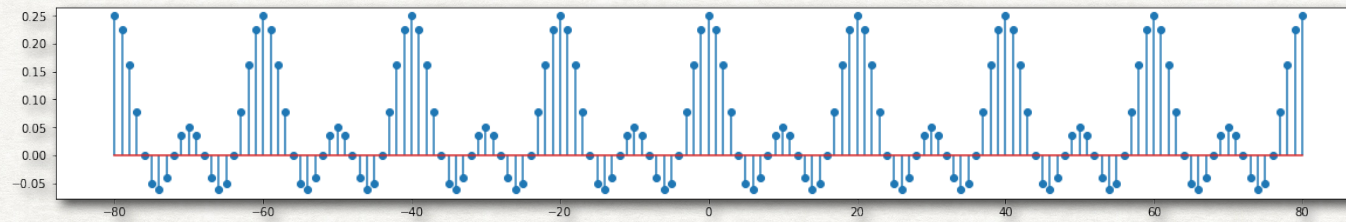
Figure 3.17 Fourier series coefficients for the periodic square wave of Example 3.12; plots of Na_k for $2N_1 + 1 = 5$ and (a) $N = 10$; (b) $N = 20$; and (c) $N = 40$.

VARYING N

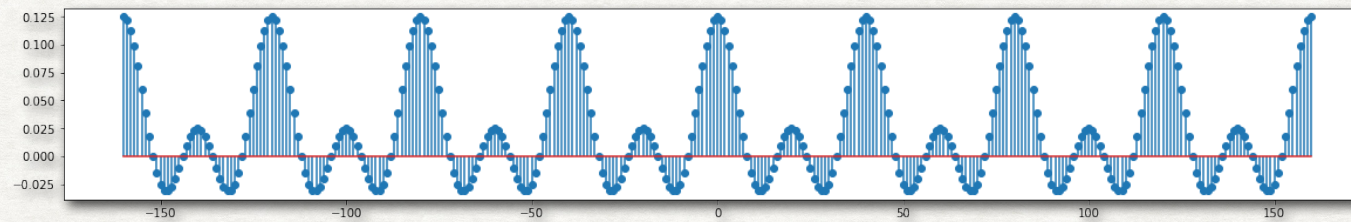
$N = 10$



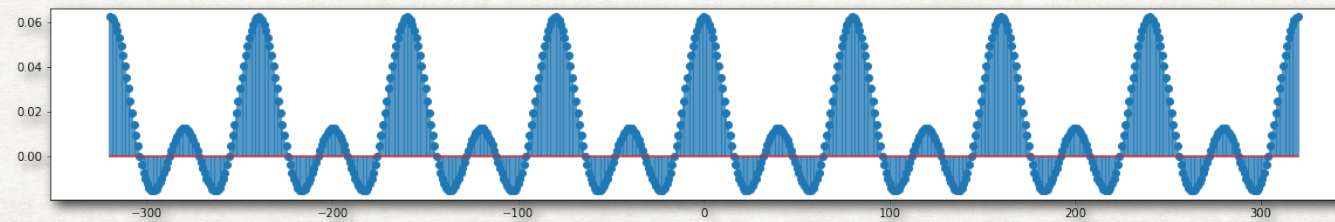
$N = 20$



$N = 40$

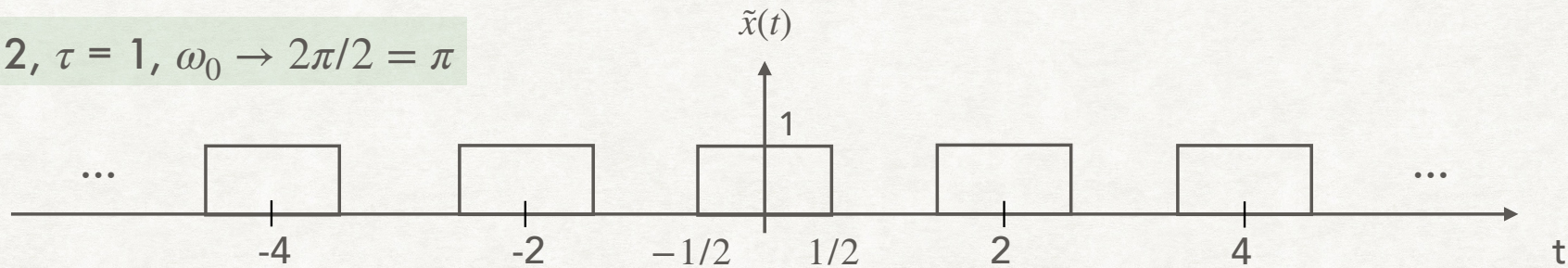


$N = 80$



(CT) FOURIER TRANSFORM

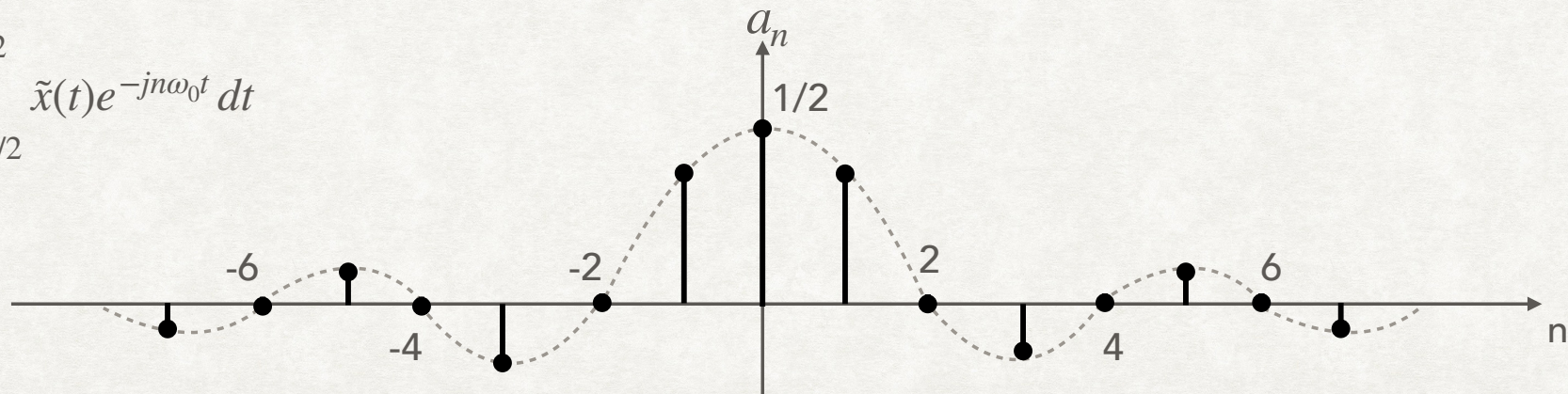
$$T = 2, \tau = 1, \omega_0 \rightarrow 2\pi/2 = \pi$$



$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn(\omega_0)t}$$

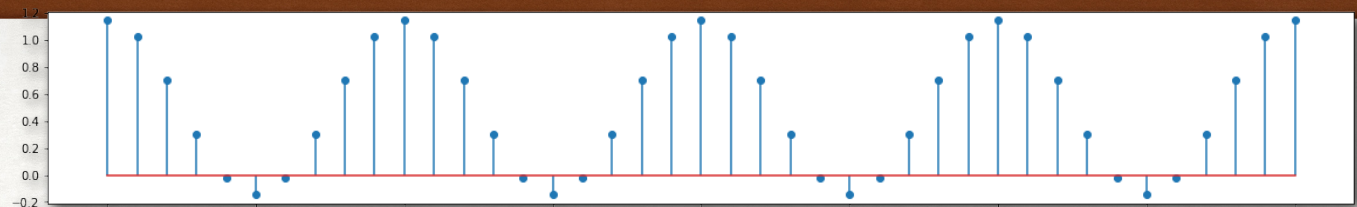
A PERIODIC SIGNAL $rect(t)$ WITH $T=2 \leftrightarrow \frac{1}{2} sinc(\frac{n}{2})$

$$a_n = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jn\omega_0 t} dt$$

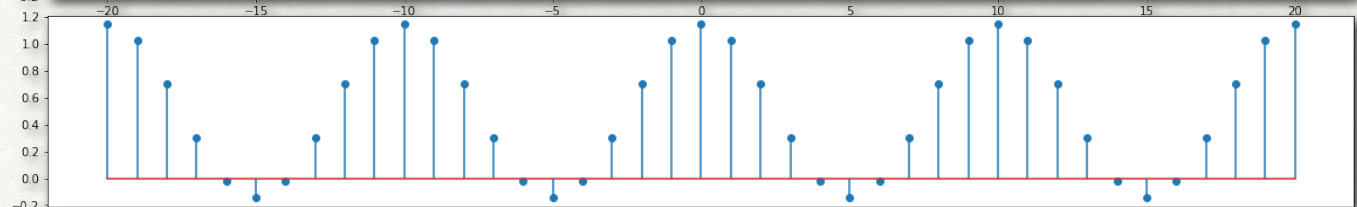


ADDING M

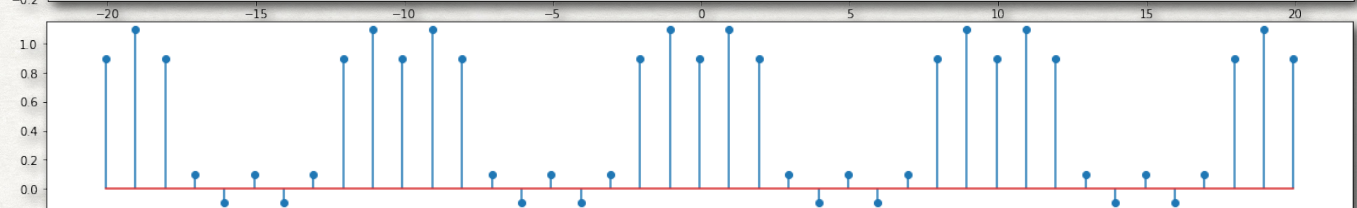
M=1



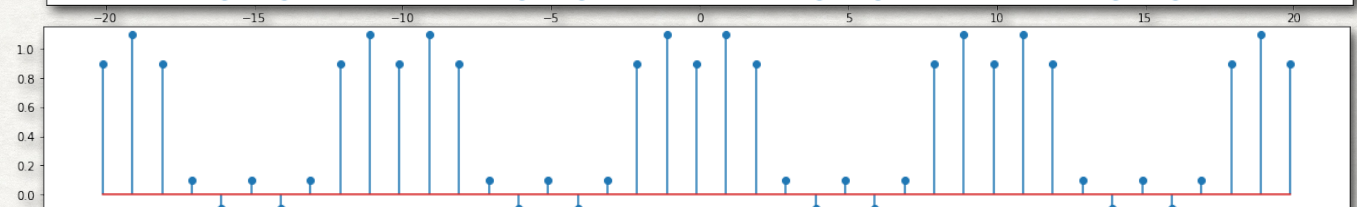
M=2



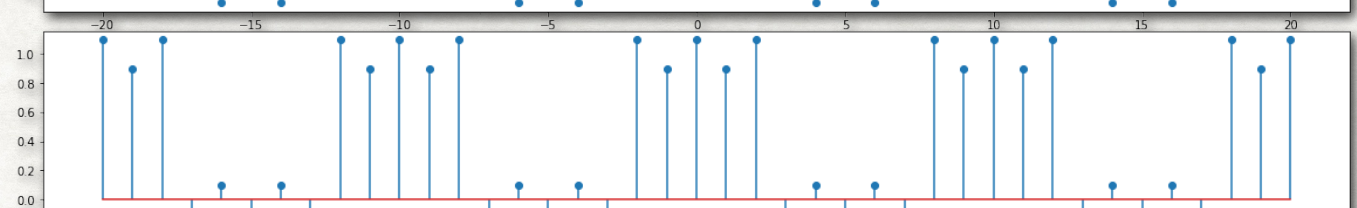
M=3



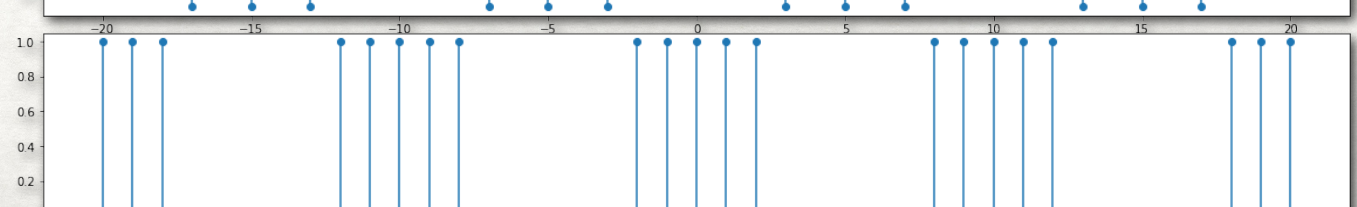
M=4



M=5



M=4 + a_5



DISCRETE-TIME FOURIER SERIES

EXERCISE #2



Considering a signal

$$x[n] = 1 + \sin\left(\frac{2\pi}{N}n\right) + 3\cos\left(\frac{2\pi}{N}n\right) + \cos\left(\frac{4\pi}{N}n + \frac{\pi}{2}\right).$$

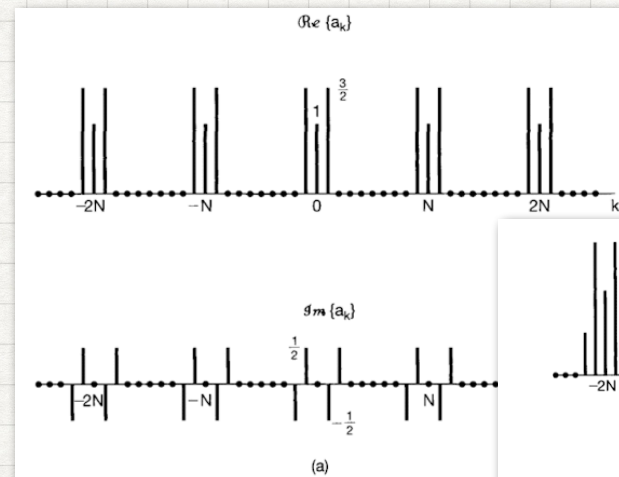
$$x[n] = 1 + \frac{1}{2j}[e^{j(2\pi/N)n} - e^{-j(2\pi/N)n}] + \frac{3}{2}[e^{j(2\pi/N)n} + e^{-j(2\pi/N)n}] + \frac{1}{2}[e^{j(4\pi/N)n + \pi/2} + e^{-j(4\pi/N)n + \pi/2}].$$

$$x[n] = 1 + \left(\frac{3}{2} + \frac{1}{2j}\right)e^{j(2\pi/N)n} + \left(\frac{3}{2} - \frac{1}{2j}\right)e^{-j(2\pi/N)n} + \left(\frac{1}{2}e^{j\pi/2}\right)e^{j2(2\pi/N)n} + \left(\frac{1}{2}e^{-j\pi/2}\right)e^{-j2(2\pi/N)n}.$$

Thus the Fourier series coefficients for this example are

$$\begin{aligned} a_0 &= 1, \\ a_1 &= \frac{3}{2} + \frac{1}{2j} = \frac{3}{2} - \frac{1}{2}j, \\ a_{-1} &= \frac{3}{2} - \frac{1}{2j} = \frac{3}{2} + \frac{1}{2}j, \\ a_2 &= \frac{1}{2}j, \\ a_{-2} &= -\frac{1}{2}j. \end{aligned}$$

For $N = 10$,



$a_{-k} = a_k^*$ for all values of k , the equality holds whenever $x[n]$ is real.

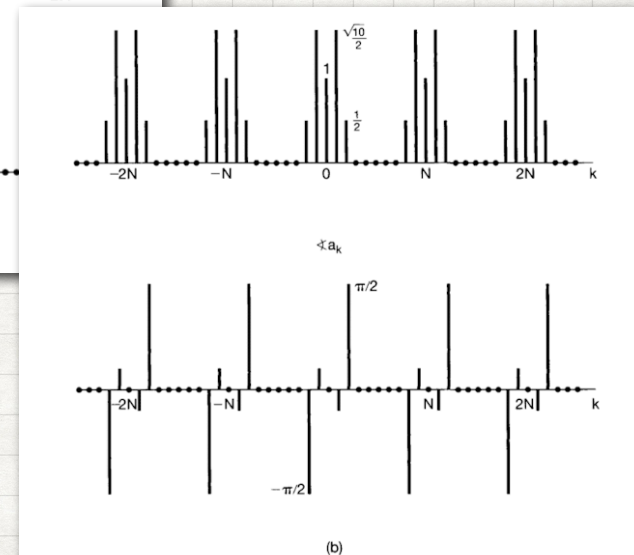
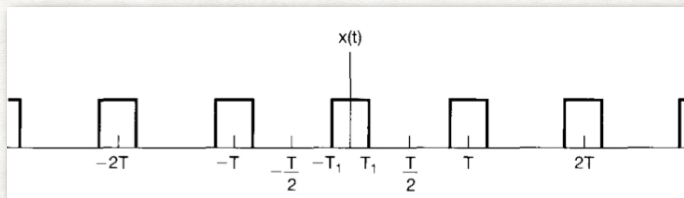


Figure 3.15 (a) Real and imaginary parts of the Fourier series coefficients in Example 3.11; (b) magnitude and phase of the same coefficients.

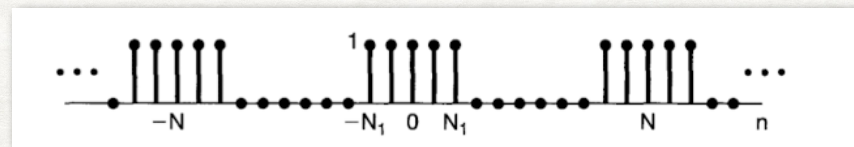
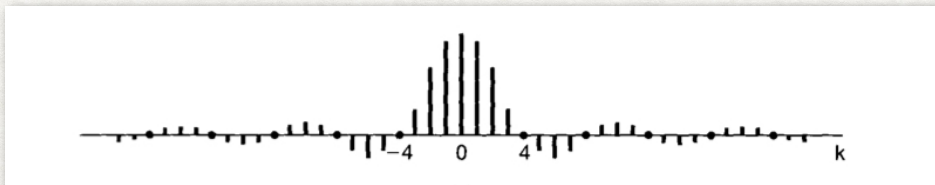
CT VS. DT COMPARISON

FOURIER SERIES



$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$$

$$a_k = \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_0^T x(t) e^{-jk(2\pi/T)t} dt$$



$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(\omega_0)n} = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}$$

